

TABLE 2—continued

Decomposition of difference in post-programme mean earnings
Bootstrapped standard errors shown in parentheses†
Percentage of mean difference attributable to components in square brackets
Earnings measured in average monthly dollars

Experimental Controls and No-shows†						Selection bias** ($\hat{B}_{Sp}$) as a % of treatment impact
	Mean difference $\hat{B}$	Non-overlap* $\hat{B}_1$	Density weighting $\hat{B}_2$	Selection bias $\hat{B}_{23}$	Average bias ($\hat{B}_{Sp}$)	
Adult males	29	13	3	38	42	97%
(std. err.) [%]	(38)	(12) [–45%]	(16) [11%]	(37) [135%]	(40)	
Adult females	9	1	–9	18	20	68%
(std. err.) [%]	(23)	(6) [9%]	(10) [–99%]	(26) [190%]	(29)	
Male youth	84	14	–21	91	99	171%
(std. err.) [%]	(31)	(9) [17%]	(11) [–25%]	(31) [108%]	(34)	
Female youth	18	3	–31	46	51	7441%
(std. err.) [%]	(24)	(16) [17%]	(16) [–170%]	(24) [254%]	(29)	

‡ They are based on 50 replications of the data with 100% sampling.

† The data used are experimental JTPA controls and experimental persons assigned to treatment who enrolled in JTPA but dropped out before receiving services. The No-show predictor model for all demographic groups includes indicator variables for site, race and recommended training services. For adult males, the model also includes earnings last year, earnings squared, an indicator for preferred language Spanish and for preferred language other than Spanish or English. For adult women, the model includes indicators for last employed 0–6 months before random assignment (RA) and for 7–12 months before RA and an indicator for whether enrollment in JTPA was required.

* A 2% trimming rule was used for adult males and females and a 5% trimming rule for youth was used in determining the overlapping support region (see Appendix C for a description of how the support is determined). The proportion of controls and no-shows falling in the overlap region (S_P) are: 91% and 91% of adult males, 90% and 92% of adult females, 91% and 93% of male youth, and 90% and 89% of female youth. A 0.06 fixed bandwidth and a biweight kernel, defined in Appendix A, were used for the nonparametric estimates.

** The final column displays the ratio of the absolute value of $\hat{B}_{Sp}$ to the absolute value of experimental impact estimate.

where $I_1^*(S_1 \setminus S_{10})$ is the set of indices in I_1^* with associated P values not in S_{10} , $I_0(S_0 \setminus S_{10})$ is the set of indices in I_0 with associated P values not in S_{10} , $I_1^*(S_{10})$ and $I_0(S_{10})$ are, respectively, the associated indices in I_1^* and I_0 with P values in S_{10} . The three terms in brackets in this decomposition are the sample analogues of the three components of B defined above. The sums are self-weighted by the $f(P|D)$ density because they average over either $D=1$ or $D=0$ values of $P(X)$.

The first panel of Table 2 reports the raw mean difference in outcomes between ENP comparison group members and controls (B), measured by monthly earnings in our study, and the contribution of each component to the overall mean bias. Y_0 corresponds to average monthly earnings over the eighteen months following random assignment or eligibility determination. For all four demographic groups, non-overlapping support and different density weighting of the P contribute most to the total bias.²⁰ After accounting for these two sources, selection bias B_3 is statistically insignificantly different from zero. This evidence suggests that evaluation methods based on the hypothesis of selection on observables eliminate the largest sources of bias, provided the evaluation parameter is estimated over the region of common support. The fifth column of numbers in the table gives the estimated average selection bias $\hat{B}_{Sp}$ over the region of overlapping support. The

20. The table footnote gives the proportion of individuals in the overlap region.

last column shows that although the bias is often small in relation to the simple mean difference, it still represents a significant fraction of the experimentally-estimated treatment impact.

Matching methods eliminate two of the three sources of bias. The bias due to nonoverlapping supports is eliminated by matching only over the region of common support. The bias due to different density weighting is eliminated because matching methods effectively reweight the nonparticipant data to equate the distribution of P . Only the bias due to differences in unobservables across groups is not eliminated.

In results reported in Heckman, Ichimura, Smith and Todd (1996a, b), we decompose the bias for each quarter following the date of random assignment or eligibility determination and show that the percentage of the total difference in quarterly means that is due to selection bias component B_3 is small for adult men, adult women and male youth relative to the other components. For female youth, it is of comparable magnitude to that of other components but all components are small in absolute value. In Heckman, Ichimura, Smith and Todd (1996a, b, c), we find that even though the average bias $\bar{B}_{s_p}$ is small, the pointwise bias $B(P)$ is not small for all four demographic groups. This evidence is contrary to what would be predicted from matching conditions (A-1), (A-3) or (A-4). It is not necessarily inconsistent with difference-in-differences assumptions (A-5) or (A-5').

10. VERIFYING THE ASSUMPTIONS THAT JUSTIFY MATCHING AND OUR EXTENSIONS OF IT

In this section, we apply methods developed in Heckman, Ichimura, Smith and Todd (1996b, c) to the JTPA data to test identifying conditions (A-3), (A-4), (A-3'), (A-4'), (A-5) and (A-5'). Assumptions (A-3) and (A-3') are decisively rejected in our data. We find some support for the weaker assumption (A-4') for one group—young males—but it is generally rejected for other groups. The weaker assumptions (A-5) and (A-5') that justify our conditional difference-in-differences method are not rejected for any group.

Specifically, we test the following hypotheses, where the subscript denotes the identifying assumption being tested:

Conditional Independence:

$$H_{(A-3)}: F(y_0|P, D=1) = F(y_0|P, D=0);$$

$$H_{(A-3')}: F(u_0|P, D=1) = F(u_0|P, D=0);$$

and

Mean Independence:

$$H_{(A-4)}: E(Y_0|P, D=1) = E(Y_0|P, D=0);$$

$$H_{(A-4')}: E(U_0|P, D=1) = E(U_0|P, D=0);$$

and

Differences in Differences Mean Independence:

$$H_{(A-5)}: E(Y_{0t} - Y_{0t'}|P, D=1) = E(Y_{0t} - Y_{0t'}|P, D=0);$$

$$H_{(A-5')}: E(U_{0t} - U_{0t'}|P, D=1) = E(U_{0t} - U_{0t'}|P, D=0),$$

where t is a post programme period and t' is a preprogramme period.

10.1. Test statistics

To test $H_{(A-3)}$ at a point $Y_0 = y_0$ conditional on $P = p$, we estimate the conditional c.d.f. $\hat{F}_d(p) = \hat{F}(y_0 | P = p, D = d)$ using local linear regression. For a given y_0 we run a nonparametric regression of the indicator variable $1(Y \leq y_0)$ on P and evaluate the function at $P = p$. $\hat{F}_d(p) - F_d(p)$ is asymptotically normally distributed, $N(B_d, V_d)$, where B_d and V_d are defined in Appendix A. Assuming that the same kernel and bandwidths are used to estimate F_1 and F_0 , under the null $H_{(A-3)}$,

$$(\hat{F}_1(p) - \hat{F}_0(p))'(\hat{V}_1 + \hat{V}_0)^{-1}(\hat{F}_1(p) - \hat{F}_0(p)) \sim \chi^2(1),$$

where $\hat{V}_1$ and $\hat{V}_0$ are consistent estimators of V_1 and V_0 that are presented in Appendix A.

To test conditional mean independence ($H_{(A-4)}$), we use local linear regression estimates of $\hat{m}_1(p) = \hat{E}(y_0 | P = p, D = 1)$ and $\hat{m}_0(p) = \hat{E}(y_0 | P = p, D = 0)$ and the fact that $(\hat{m}_d(p) - m_d(p)) \sim N(\tilde{B}_d, \tilde{V}_d)$ where $\tilde{B}_d$ and $\tilde{V}_d$ are defined in Appendix A. Assuming the same kernel and bandwidths are used to estimate m_0 and m_1 , under the null

$$(\hat{m}_1(p) - \hat{m}_0(p))'(\tilde{V}_1 + \tilde{V}_0)^{-1}(\hat{m}_1(p) - \hat{m}_0(p)) \sim \chi^2_1,$$

where $\tilde{V}_1$ and $\tilde{V}_0$ are feasible covariance estimators for $\hat{m}_0$ and $\hat{m}_1$ defined in Appendix A.

An important advantage of using local linear regression methods instead of ordinary kernel regression methods in implementing these tests is that, under the two nulls, the bias terms for F_1 and F_0 , (B_1, B_0) , and for m_1 and m_0 , $(\tilde{B}_1, \tilde{B}_0)$, cancel in the test statistics when the same bandwidth is used to estimate $(\hat{F}_1(p), \hat{F}_0(p))$ and $(\hat{m}_1(p), \hat{m}_0(p))$ respectively. With ordinary kernel methods, the bias depends on the data density $f_d(p)$, so the bias terms do not cancel and the test statistic has a noncentral chi-squared distribution. (Heckman, Ichimura, Smith and Todd (1996b, c).)

Tests of $H_{(A-3')}$ and $H_{(A-4')}$ replace Y_0 by the corresponding U_0 residuals. Appendix A defines the test statistic for the second hypothesis, $H_{(A-4')}$, gives estimators of the variances needed to perform the tests and generalizes the test statistics to apply to panel data. Tests of $H_{(A-5)}$ and $H_{(A-5')}$ are constructed on similar principles. The test for $H_{(A-5)}$ is just a version of the test for $H_{(A-4)}$ using differences in the conditional means as input to a $\chi^2(1)$ test. Residuals replace conditional means of outcomes in the test of $H_{(A-5')}$. Details on how to construct these tests are given in Appendix A.

10.2. Empirical evidence on the validity of matching assumptions

In tables available on request from the authors, we test the conventional identifying assumption $H_{(A-3)}$ and the corresponding assumption for population disturbances $H_{(A-3')}$. The test is performed at selected quantiles of the outcome distribution, at different values of P , and overall. Separate tests are conducted for pre-programme and post-random assignment time periods. For adult men, $H_{(A-3)}$ is decisively rejected. For adult women $H_{(A-3)}$ is rejected at some quantiles of P . For male youth, $H_{(A-3)}$ is not rejected, while for female youth there are a few rejections of the hypothesis. In tests of $H_{(A-3')}$, we reject the hypothesis for adult men and women and for female youth but not for male youth.

Table 3(a) presents tests results for the weaker hypothesis of conditional mean independence of Y_0 ($H_{(A-4)}$) and U_0 ($H_{(A-4')}$) that still justify matching to recover means. Since conditional independence implies mean independence and mean independence suffices for our parameter of interest, this is the central hypothesis for testing the validity of matching in our samples. For all four demographic groups, in the pre-programme period we reject

TABLE 3(a)

P-values from tests for conditional mean independence $(H_{(A-4)}): E(Y_0 | D=1, P) = E(Y_0 | D=0, P)$ $(H_{(A-4')}): E(U_0 | D=1, P) = E(U_0 | D=0, P)$

<i>P</i> -points	Adult men				Adult women			
	Earnings ($H_{(A-4)}$)		Residuals ($H_{(A-4')}$)		Earnings ($H_{(A-4)}$)		Residuals ($H_{(A-4')}$)	
	Pre-programme quarters†	Post-programme quarters†	Pre-programme quarters†	Post-programme quarters†	Pre-programme quarters†	Post-programme quarters†	Pre-programme quarters†	Post-programme quarters†
0.0025	0.0242	0.0000	0.0293	0.0002	0.0751	0.0029	0.1472	0.0363
0.005	0.0803	0.0002	0.0815	0.0004	0.1671	0.0057	0.1847	0.0298
0.01	0.2416	0.0042	0.2586	0.0040	0.3963	0.0104	0.3035	0.0140
0.02	0.1919	0.1224	0.4078	0.2056	0.3609	0.0172	0.3836	0.0180
0.03	0.1238	0.4363	0.5680	0.6563	0.0577	0.0677	0.1048	0.1243
0.04	0.1585	0.7659	0.7177	0.9060	0.0145	0.1418	0.0790	0.2570
0.05	0.3271	0.9423	0.8064	0.9885	0.0402	0.2052	0.2718	0.4510
0.10	0.8464	0.1678	0.7456	0.2591	0.0702	0.4406	0.4752	0.8423
Overall	0.0159	0.0000	0.2515	0.0001	0.0006	0.0000	0.0375	0.0008

<i>P</i> -points	Male youth				Female youth			
	Earnings ($H_{(A-4)}$)		Residuals ($H_{(A-4')}$)		Earnings ($H_{(A-4)}$)		Residuals ($H_{(A-4')}$)	
	Pre-programme quarters†	Post-programme quarters†	Pre-programme quarters†	Post-programme quarters†	Pre-programme quarters†	Post-programme quarters†	Pre-programme quarters†	Post-programme quarters†
0.0025	0.2277	0.0333	0.3877	0.2142	0.1323	0.0076	0.1418	0.3047
0.005	0.2964	0.0997	0.3547	0.2911	0.0906	0.0166	0.0770	0.3648
0.01	0.3966	0.3047	0.3373	0.4894	0.0490	0.0697	0.0274	0.5512
0.02	0.5400	0.3426	0.4702	0.6166	0.0273	0.1853	0.0107	0.4803
0.03	0.7053	0.1845	0.6913	0.5214	0.0540	0.0621	0.0277	0.1570
0.04	0.7095	0.1239	0.7342	0.4567	0.1948	0.0290	0.2278	0.0575
0.05	0.4260	0.1329	0.5988	0.4315	0.4551	0.0236	0.6279	0.0245
0.10	0.0000	0.9124	0.0000	0.1944	0.1186	0.0441	0.2412	0.0000
Overall	0.0147	0.0251	0.0015	0.3561	0.0009	0.0000	0.0006	0.0001

* A fixed bandwidth of 0.06 and a biweight kernel, defined in Appendix A, are used in the test. The models for the probability of participation, P , are described in the footnotes to Table 2.

† Tests are performed jointly across pre-programme quarters $t = -1$ to $t = -6$ or post-programme quarters $t = 1$ to $t = 6$.

TABLE 3(b)

P-values from tests for difference-in-differences† $(H_{(A-5)}): E(Y_{0t} - Y_{0t'} | D=1, P) = E(Y_{0t} - Y_{0t'} | D=0, P)$ $(H_{(A-5')}): E(U_{0t} - U_{0t'} | D=1, P) = E(U_{0t} - U_{0t'} | D=0, P)$ (Null tested jointly over $t \in \{1, 2, 3, 4, 5, 6\}$)

<i>P</i> -points	Adult males		Adult females		Male youth		Female youth	
	Earnings	Residuals	Earnings	Residuals	Earnings	Residuals	Earnings	Residuals
0.0025	0.0782	0.0937	0.4139	0.2340	0.7178	0.7079	0.0320	0.0707
0.005	0.0885	0.0966	0.2924	0.1454	0.7255	0.7268	0.0269	0.0538
0.01	0.1377	0.1184	0.1195	0.0482	0.6878	0.7404	0.0377	0.0596
0.02	0.3946	0.2865	0.0865	0.0454	0.6594	0.7968	0.1702	0.2420
0.03	0.7149	0.6007	0.4690	0.3661	0.6548	0.8426	0.4669	0.6340
0.04	0.9167	0.8745	0.7672	0.6089	0.6836	0.8617	0.7446	0.8577
0.05	0.9727	0.9506	0.6091	0.4626	0.7717	0.8565	0.6632	0.7594
0.10	0.9677	0.9833	0.7386	0.5098	0.8346	0.4245	0.9813	0.9786
Overall	0.5260	0.4403	0.3364	0.0506	0.9832	0.9910	0.0533	0.2119

† A fixed bandwidth equal to 0.06 and a biweight kernel, defined in Appendix A, are used. The models for the probability of participation, P , are those given in the footnote to Table 2. The test is for symmetric differences around $t=0$, the date of enrollment into the programme, $t = -t'$.